

B.Tech First Year | Engineering Physics

Engineering Physics

Complete Study Module

UNIT V

Semiconductors

Topics: Energy Bands · Intrinsic & Extrinsic Semiconductors · Carrier Transport · Hall Effect

Textbook References:

M.N. Avadhanulu et al. – S. Chand (2019) | D.K. Bhattacharya – Oxford (2015)

Exam-Oriented · Derivation-Ready · Diagram-Rich

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1. Introduction to Semiconductors

1.1 Definition

A **semiconductor** is a material whose electrical conductivity lies between that of a conductor and an insulator at room temperature. Semiconductors have a resistivity in the range of 10^{-4} to $0.5 \text{ } \Omega \cdot \text{m}$ and possess a narrow forbidden energy gap ($E_g \approx 0.1 - 3 \text{ eV}$).

1.2 Why Study Semiconductors?

Semiconductors form the backbone of modern electronics. Devices such as diodes, transistors, solar cells, LEDs, and integrated circuits are all fabricated from semiconductor materials. Understanding their physics is essential to engineering innovation.

1.3 Common Semiconductor Materials

Material	Symbol	Band Gap (eV)	Type
Silicon	Si	1.12	Elemental
Germanium	Ge	0.66	Elemental
Gallium Arsenide	GaAs	1.43	Compound
Indium Phosphide	InP	1.35	Compound
Gallium Nitride	GaN	3.4	Compound

1.4 Key Characteristics

- Conductivity increases with temperature (negative temperature coefficient).
- Can be doped with impurities to alter electrical properties drastically.
- Charge is carried by both electrons and holes.
- Exhibit photoconductivity, thermoelectric effects, and rectification.

Note: At 0 K, a pure semiconductor behaves like a perfect insulator because no free carriers exist. As temperature increases, covalent bonds break thermally, generating electron-hole pairs.

2. Energy Band Theory

2.1 Formation of Energy Bands

In an isolated atom, electrons occupy discrete energy levels. When N atoms are brought together to form a crystal, the Pauli exclusion principle forbids two electrons from having the same quantum state. Consequently, each discrete atomic level splits into N closely spaced sub-levels, forming an almost continuous **energy band**.

Step 1: Isolated atom: electrons in sharp energy levels (e.g., 1s, 2s, 2p, ...).

Step 2: As atoms approach each other, electron wavefunctions overlap.

Step 3: Pauli exclusion: identical states are forbidden $\rightarrow$ each level splits into N sub-levels.

Step 4: For $N \sim 10^{23}$ atoms/cm³, sub-levels are so close they form a continuous band.

Step 5: Between bands, forbidden energy gaps (band gaps) exist where no electron state is allowed.

2.2 Important Energy Bands

- **Valence Band (VB):** Highest filled (or partially filled) energy band at 0 K. Contains valence electrons.
- **Conduction Band (CB):** Next higher energy band, partially or fully empty. Free electrons reside here.
- **Forbidden Gap (E_g):** Energy gap between top of VB and bottom of CB. No allowed electron states exist here.

Band Gap Energy:

$$E_g = E_C - E_V$$

where E_C = bottom of conduction band, E_V = top of valence band

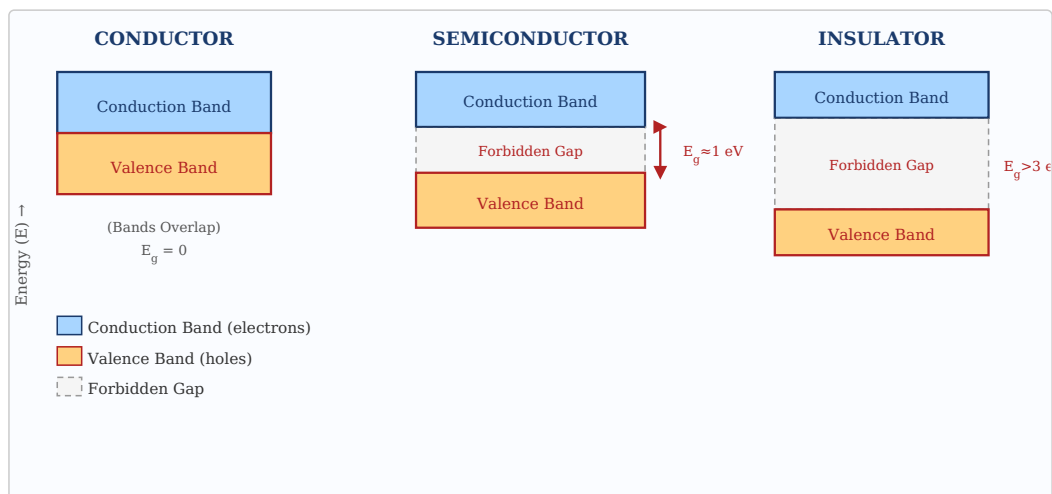


Fig. 2.1 - Energy Band Diagrams for Conductor, Semiconductor, and Insulator

2.3 Kronig-Penney Model (Brief)

The Kronig-Penney model uses a periodic square-well potential to represent the crystal lattice. Solving Schrödinger's equation in this potential shows that electron energies cluster into allowed bands separated by forbidden gaps. The width of the gap depends on the height and width of the potential barriers (i.e., the nature of the crystal).

Key Result (Kronig-Penney):

$$\cos(ka) = f(E) \quad \text{where } k = \text{wave vector, } a = \text{lattice constant}$$

Solutions exist only for certain energy ranges $\rightarrow$ allowed bands and forbidden gaps.

3. Classification of Crystalline Solids

3.1 Conductors

In conductors (metals), the conduction band and valence band **overlap** at room temperature. A large number of free electrons ($\sim 10^{28}/\text{m}^3$) are always available for conduction. Resistivity is very low ($10^{-8} - 10^{-6} \Omega\cdot\text{m}$). Example: Copper, Aluminium, Silver.

3.2 Insulators

Insulators have a completely filled valence band and a completely empty conduction band, separated by a very large forbidden gap ($E_g > 3 \text{ eV}$). Electrons cannot jump from VB to CB at normal temperatures. Example: Diamond ($E_g = 5.4 \text{ eV}$), SiO_2 ($E_g \approx 9 \text{ eV}$).

3.3 Semiconductors

Semiconductors have a narrow forbidden gap ($0.1 - 3 \text{ eV}$). At 0 K, they behave like insulators. At room temperature, thermal energy causes some electrons to jump from VB to CB, leaving behind **holes** in the VB, thereby enabling conduction.

Table 3.1 - Comparative Properties of Conductors, Semiconductors, and Insulators

Property	Conductor	Semiconductor	Insulator
Band gap (E_g)	0 (overlap)	0.1 - 3 eV	> 3 eV
Resistivity ($\Omega\cdot\text{m}$)	$10^{-8} - 10^{-6}$	$10^{-4} - 0.5$	$10^8 - 10^{16}$
Free carrier density	$\sim 10^{28} / \text{m}^3$	$10^{10} - 10^{17} / \text{m}^3$	Negligible
Effect of temperature	Resistance $\uparrow$	Resistance $\downarrow$	Slight $\downarrow$
Effect of doping	Not significant	Large increase in σ	Not significant
Example	Cu, Ag, Al	Si, Ge, GaAs	Diamond, Glass

4. Intrinsic Semiconductors

4.1 Definition

An **intrinsic semiconductor** is a pure semiconductor without any impurity atoms. Its electrical properties are determined solely by its crystal structure and temperature. In an intrinsic semiconductor, the number of electrons in the conduction band equals the number of holes in the valence band.

Intrinsic condition:

$$n = p = n_i$$

where n = electron concentration, p = hole concentration, n_i = intrinsic carrier concentration

4.2 Bond Model of Intrinsic Semiconductor

Silicon has 4 valence electrons. In the crystal, each atom forms 4 covalent bonds with neighbouring atoms. At 0 K, all bonds are intact - no free carriers. At finite temperature, thermal vibrations break some covalent bonds, releasing an electron into the conduction band and creating a hole in the valence band. This process is called **thermal generation of electron-hole pairs (EHP)**.

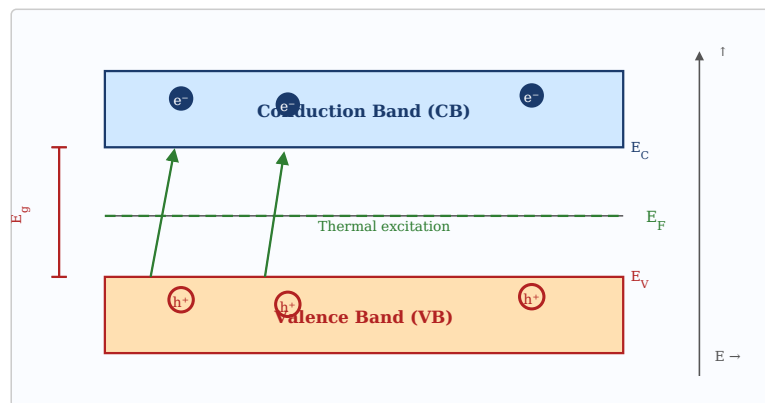


Fig. 4.1 - Band diagram of Intrinsic Semiconductor (Fermi level lies in the middle of the gap)

4.3 Fermi Level in Intrinsic Semiconductor

The **Fermi level (E_F)** is defined as the energy level at which the probability of finding an electron is exactly 0.5 (50%). For an intrinsic semiconductor, the Fermi level lies approximately at the **middle of the forbidden gap**.

Fermi level in Intrinsic Semiconductor:

$$E_F = (E_C + E_V) / 2 + (kT/2) \cdot \ln(N_V/N_C)$$

$$\text{At } T = 0 \text{ K and when } m_e^* = m_h^*: \quad E_F = (E_C + E_V) / 2 = E_V + E_g/2$$

5. Charge Carrier Density in Intrinsic Semiconductors

5.1 Fermi-Dirac Distribution

The probability that an energy state E is occupied by an electron is given by the Fermi-Dirac distribution function:

$$f(E) = 1 / [1 + \exp((E - E_F) / kT)]$$

where k = Boltzmann constant (1.38×10^{-23} J/K), T = absolute temperature, E_F = Fermi energy

5.2 Density of States

The number of allowed quantum states per unit volume per unit energy range near the bottom of the conduction band is:

$$g_c(E) = (4\pi / h^3) \cdot (2m_e^*)^{3/2} \cdot (E - E_c)^{1/2}$$

Similarly for valence band: $g_v(E) = (4\pi / h^3) \cdot (2m_h^*)^{3/2} \cdot (E_v - E)^{1/2}$

5.3 Electron Concentration in CB

Derivation:

The number of electrons per unit volume in the conduction band:

$$n = \int_{E_c}^{\infty} g_c(E) \cdot f(E) dE$$

For $(E - E_F) \gg kT$ (non-degenerate approximation), $f(E) \approx \exp(-(E - E_F)/kT)$

Substituting and integrating:

$$n = N_c \cdot \exp(-(E_c - E_F) / kT)$$

where the **effective density of states in CB**:

$$N_c = 2 \cdot (2\pi m_e^* kT / h^2)^{3/2}$$

5.4 Hole Concentration in VB

Similarly:

$$p = N_v \cdot \exp(-(E_F - E_v) / kT)$$

where $N_v = 2 \cdot (2\pi m_h^* kT / h^2)^{3/2}$ = effective density of states in VB

5.5 Intrinsic Carrier Concentration (n_i)

For intrinsic semiconductor: $n = p = n_i$

Taking the product: $n \cdot p = n_i^2 = N_c \cdot N_v \cdot \exp(-E_g/kT)$

Therefore:

Intrinsic Carrier Concentration (most important formula):

$$n_i = \sqrt{(N_c \cdot N_v) \cdot \exp(-E_g / 2kT)}$$

$$n_i = A \cdot T^{3/2} \cdot \exp(-E_g / 2kT)$$

For Si at 300 K: $n_i \approx 1.5 \times 10^{10} \text{ cm}^{-3}$

For Ge at 300 K: $n_i \approx 2.4 \times 10^{13} \text{ cm}^{-3}$

Law of Mass Action: $n \cdot p = n_i^2$ — This holds for both intrinsic and extrinsic semiconductors in thermal equilibrium.

6. Electrical Conductivity of Semiconductors

6.1 Charge Carrier Mobility

Mobility (μ) is defined as the drift velocity acquired by a charge carrier per unit electric field.

$\mu = v_d / E$ (units: $\text{m}^2/\text{V}\cdot\text{s}$ or $\text{cm}^2/\text{V}\cdot\text{s}$)
 μ_n = mobility of electrons | μ_p = mobility of holes
For Si: $\mu_n \approx 1350 \text{ cm}^2/\text{V}\cdot\text{s}$ | $\mu_p \approx 480 \text{ cm}^2/\text{V}\cdot\text{s}$
For Ge: $\mu_n \approx 3900 \text{ cm}^2/\text{V}\cdot\text{s}$ | $\mu_p \approx 1900 \text{ cm}^2/\text{V}\cdot\text{s}$

6.2 Derivation of Electrical Conductivity

Current density due to electrons: $J_n = n \cdot q \cdot v_{dn} = n \cdot q \cdot \mu_n \cdot E$

Current density due to holes: $J_p = p \cdot q \cdot v_{dp} = p \cdot q \cdot \mu_p \cdot E$

Total current density: $J = J_n + J_p = (n \cdot q \cdot \mu_n + p \cdot q \cdot \mu_p) \cdot E$

Since $J = \sigma \cdot E$:

Electrical Conductivity:

$$\sigma = q \cdot (n \cdot \mu_n + p \cdot \mu_p)$$

For intrinsic semiconductor ($n = p = n_i$):

$$\sigma_i = n_i \cdot q \cdot (\mu_n + \mu_p)$$

$$\sigma_i = q \cdot (\mu_n + \mu_p) \cdot A \cdot T^{3/2} \cdot \exp(-E_g/2kT)$$

6.3 Temperature Dependence of Conductivity

As temperature increases, n_i increases exponentially (dominant effect), so conductivity of a semiconductor **increases with temperature**. This is opposite to metals.

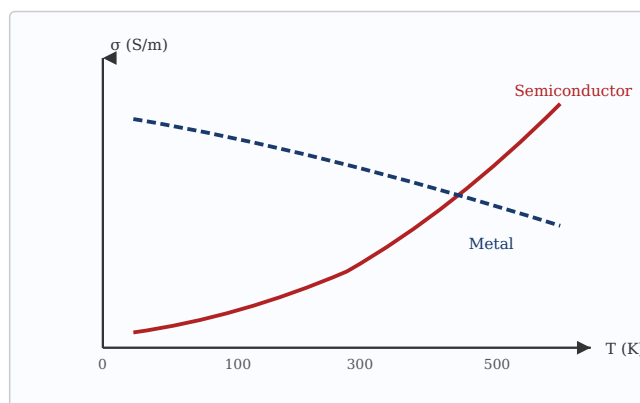


Fig. 6.1 – Conductivity vs Temperature: Semiconductor (↑) vs Metal (↓)

Table 6.1 – Carrier Mobility of Common Semiconductors at 300 K

Material	$\mu_n \text{ (cm}^2/\text{V}\cdot\text{s)}$	$\mu_p \text{ (cm}^2/\text{V}\cdot\text{s)}$	$E_g \text{ (eV)}$	$n_i \text{ (cm}^{-3}\text{)}$
Silicon (Si)	1350	480	1.12	1.5×10^{10}
Germanium (Ge)	3900	1900	0.66	2.4×10^{13}

GaAs	8500	400	1.43	1.8×10^6
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7. Extrinsic Semiconductors

7.1 Definition & Need for Doping

An **extrinsic semiconductor** is a semiconductor in which impurity atoms (dopants) are intentionally added to the pure (intrinsic) crystal to drastically alter its electrical conductivity. The process of adding impurities is called **doping**.

7.2 n-type Semiconductor

When a **pentavalent impurity** (5 valence electrons, e.g., Phosphorus, Arsenic, Antimony) is added to Si or Ge:

- The impurity atom uses 4 electrons to form covalent bonds with Si neighbours.
- The 5th electron is loosely bound and easily donated to the CB.
- These impurities are called **donor impurities** (donor level E_D just below E_C).
- Majority carriers = electrons; Minority carriers = holes.

n-type carrier concentrations:

$$n \approx N_D \text{ (donor concentration) [for } N_D \gg n_i]$$

$$p = n_i^2 / N_D \text{ (minority holes)}$$

7.3 p-type Semiconductor

When a **trivalent impurity** (3 valence electrons, e.g., Boron, Aluminium, Gallium) is added:

- The impurity atom has only 3 electrons → creates a vacancy (hole) in the covalent network.
- This hole can accept an electron from a neighbouring bond → **acceptor impurity**.
- Acceptor level E_A is just above E_V .
- Majority carriers = holes; Minority carriers = electrons.

p-type carrier concentrations:

$$p \approx N_A \text{ (acceptor concentration) [for } N_A \gg n_i]$$

$$n = n_i^2 / N_A \text{ (minority electrons)}$$

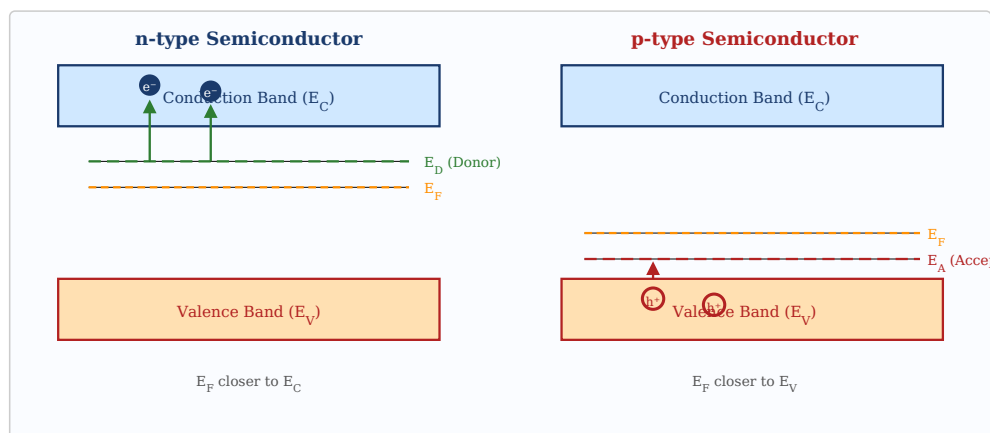


Fig. 7.1 – Band diagrams of n-type (left) and p-type (right) extrinsic semiconductors

Table 7.1 – Comparison: n-type vs p-type Semiconductors

Property	n-type	p-type
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Dopant	Pentavalent (P, As, Sb)	Trivalent (B, Al, Ga)
Majority carrier	Electrons (e^-)	Holes (h^+)
Minority carrier	Holes (h^+)	Electrons (e^-)
Fermi level	Close to E_C	Close to E_V
Impurity level	Donor level (E_D) below E_C	Acceptor level (E_A) above E_V
Charge neutrality	$n \approx N_D$	$p \approx N_A$
Hall coefficient sign	Negative ($R_H < 0$)	Positive ($R_H > 0$)

Table 7.2 - Intrinsic vs Extrinsic Semiconductors

Feature	Intrinsic	Extrinsic
Purity	Pure (undoped)	Doped with impurities
Carrier generation	Thermal (EHP generation)	Both thermal + ionization of dopants
n vs p	$n = p = n_i$	$n \neq p$
Conductivity control	Only by temperature	By doping concentration
Applications	Thermistors, basic study	Diodes, transistors, ICs

8. Fermi Level Concepts

8.1 Fermi Level in n-type Semiconductor

As donor concentration N_D increases, the Fermi level moves **closer to the conduction band**.

Fermi Level in n-type:

$$E_F = E_C - kT \cdot \ln(N_C / N_D)$$

or equivalently: $E_F - E_i = kT \cdot \ln(N_D / n_i)$

where E_i = intrinsic Fermi level (midgap)

8.2 Fermi Level in p-type Semiconductor

As acceptor concentration N_A increases, the Fermi level moves **closer to the valence band**.

Fermi Level in p-type:

$$E_F = E_V + kT \cdot \ln(N_V / N_A)$$

or: $E_i - E_F = kT \cdot \ln(N_A / n_i)$

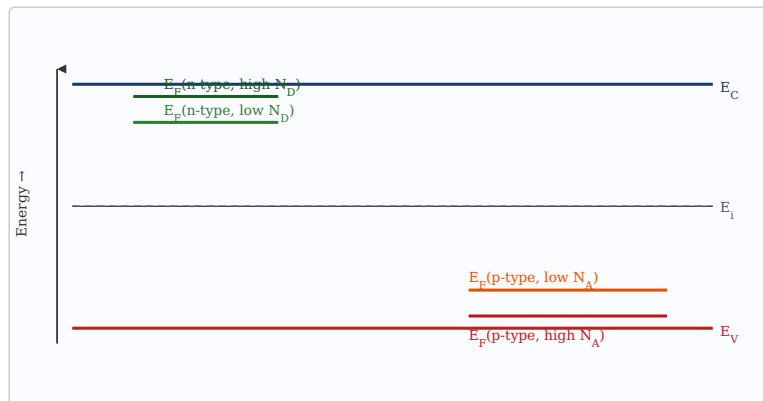


Fig. 8.1 – Fermi Level positions for intrinsic, n-type, and p-type semiconductors

8.3 Carrier Concentration Dependence on Doping

The carrier concentration varies exponentially with doping level and inversely with temperature through the Fermi-Dirac statistics. At very heavy doping (degenerate semiconductor), E_F enters the conduction band (n-type) or valence band (p-type).

9. Temperature Dependence of Fermi Level

9.1 Effect in n-type Semiconductor

At very low temperature ($T \rightarrow 0$ K): only a few donors are ionized; E_F lies **between E_D and E_C** . As temperature rises, more donors are ionized, E_F moves slightly downward. At high temperature (intrinsic regime), the thermally generated carriers dominate and E_F moves back toward midgap (E_i).

Temperature behaviour of E_F in n-type:

At $T = 0$ K: $E_F = (E_C + E_D) / 2$

At moderate T: $E_F = E_C - kT \cdot \ln(N_C/N_D)$

At very high T: $E_F \rightarrow E_i$ (midgap)

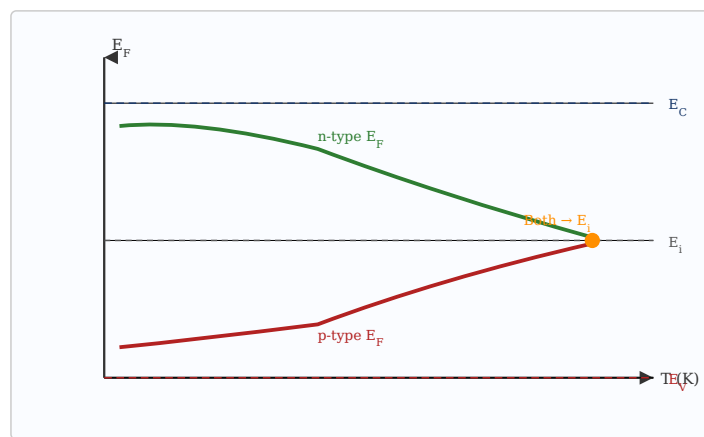


Fig. 9.1 - Fermi level vs Temperature for n-type and p-type semiconductors

9.2 Effect in p-type Semiconductor

At very low temperature, E_F lies **between E_A and E_V** . As temperature increases, more acceptors are ionized, and eventually at high temperatures, thermally generated EHPs dominate, driving E_F back toward E_i .

Key Observation: At very high temperatures, both n-type and p-type extrinsic semiconductors become effectively intrinsic because thermally generated carriers overwhelm the doped carriers. The Fermi level in both cases converges to the intrinsic Fermi level E_i .

10. Drift Current

10.1 Definition

Drift current is the electric current produced in a semiconductor due to the motion of charge carriers (electrons and holes) under the influence of an externally applied electric field. The carriers move with an average **drift velocity** proportional to the applied field.

10.2 Theory & Derivation

When an electric field E is applied, electrons experience a force $F = -qE$ (opposite to E) and holes experience $F = +qE$ (along E direction).

Drift velocity of electrons: $v_{dn} = -\mu_n \cdot E$

Drift velocity of holes: $v_{dp} = +\mu_p \cdot E$

Current density due to electron drift: $J_{n,drift} = (-q) \cdot n \cdot v_{dn} = n \cdot q \cdot \mu_n \cdot E$

Current density due to hole drift: $J_{p,drift} = q \cdot p \cdot v_{dp} = p \cdot q \cdot \mu_p \cdot E$

Total drift current density:

Drift Current Density:

$$J_{drift} = J_{n,drift} + J_{p,drift}$$

$$J_{drift} = q \cdot (n \cdot \mu_n + p \cdot \mu_p) \cdot E$$

$$J_{drift} = \sigma \cdot E \quad (\text{Ohm's Law in microscopic form})$$

$$\text{where } \sigma = q \cdot (n \cdot \mu_n + p \cdot \mu_p) = \text{conductivity}$$

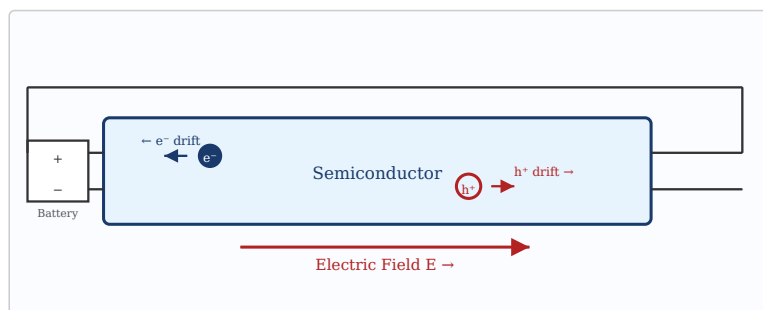


Fig. 10.1 – Drift Current: Electrons drift opposite to E ; Holes drift along E

10.3 Key Points

- Drift current is caused by applied electric field (external source).
- Electron and hole drift currents flow in the **same direction** (conventional current from + to -).
- Drift current is proportional to carrier concentration and mobility.
- Dominant in uniformly doped semiconductors under an applied field.

11. Diffusion Current

11.1 Definition

Diffusion current arises due to the non-uniform concentration of charge carriers in a semiconductor. Carriers spontaneously diffuse from regions of **high concentration** to regions of **low concentration**, even in the absence of an electric field.

11.2 Fick's First Law of Diffusion

The particle flux (number of particles crossing unit area per second) is proportional to the concentration gradient:

$$\begin{aligned}\Phi_n &= -D_n \cdot (dn/dx) && \text{(electron flux)} \\ \Phi_p &= -D_p \cdot (dp/dx) && \text{(hole flux)} \\ \text{where } D_n, D_p &= \text{diffusion coefficients (cm}^2/\text{s)}\end{aligned}$$

11.3 Diffusion Current Density

Since electrons carry charge $-q$:

$$J_{n,\text{diff}} = q \cdot D_n \cdot (dn/dx)$$

Since holes carry charge $+q$:

$$J_{p,\text{diff}} = -q \cdot D_p \cdot (dp/dx)$$

Diffusion Current Density:

$$J_{n,\text{diff}} = q \cdot D_n \cdot (dn/dx)$$

$$J_{p,\text{diff}} = -q \cdot D_p \cdot (dp/dx)$$

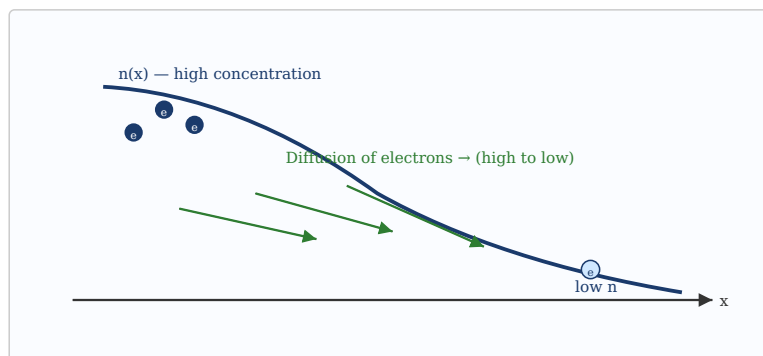


Fig. 11.1 - Diffusion Current: Electrons diffuse from high to low concentration

11.4 Total Current Density

In general, both drift and diffusion currents coexist. The total current density for each carrier type is:

Total Current Density (Drift + Diffusion):

$$J_n = q \cdot n \cdot \mu_n \cdot E + q \cdot D_n \cdot (dn/dx)$$

$$J_p = q \cdot p \cdot \mu_p \cdot E - q \cdot D_p \cdot (dp/dx)$$

$$J_{\text{total}} = J_n + J_p$$

Table 11.1 – Comparison: Drift Current vs Diffusion Current

Feature	Drift Current	Diffusion Current
Cause	Applied electric field (E)	Concentration gradient (dn/dx)
Direction	Along E (holes) / Against E (electrons)	High → Low concentration
Formula	$J = \sigma E = q(n\mu_n + p\mu_p)E$	$J_n = qD_n dn/dx$
Requires E field?	Yes	No
Requires gradient?	No	Yes
Occurs in	Homogeneous semiconductors	p-n junctions, LEDs, solar cells

12. Einstein's Relation & Hall Effect

12.1 Einstein's Relation

Einstein's relation connects the diffusion coefficient D with the mobility μ of charge carriers. It is derived by considering that in thermal equilibrium, the drift and diffusion currents must exactly cancel each other.

12.1.1 Derivation

Setup: Consider a non-uniformly doped n-type semiconductor in thermal equilibrium (no net current).

The built-in electric field due to the doping gradient is E , and:

$J_{n,\text{total}} = 0$ (thermal equilibrium)

$$q \cdot n \cdot \mu_n \cdot E + q \cdot D_n \cdot (dn/dx) = 0 \quad \dots(1)$$

In thermal equilibrium, the electron concentration follows:

$$n(x) = n_0 \cdot \exp(q\phi(x)/kT) \quad \text{where } \phi \text{ is the electrostatic potential}$$

$$\text{Differentiating: } dn/dx = n \cdot (q/kT) \cdot (d\phi/dx) = n \cdot (q/kT) \cdot (-E) = -nqE/kT \quad \dots(2)$$

Substituting (2) into (1):

$$q \cdot n \cdot \mu_n \cdot E + q \cdot D_n \cdot (-nqE/kT) = 0$$

$$\mu_n - D_n \cdot q/kT = 0$$

Einstein's Relation (fundamental result):

$$D_n / \mu_n = kT / q = V_T$$

$$D_p / \mu_p = kT / q = V_T$$

where $V_T = kT/q = \text{Thermal voltage} \approx 26 \text{ mV at } 300 \text{ K}$

Also written as: $D = \mu \cdot (kT/q)$

Physical Meaning: Einstein's relation shows that diffusion and drift are not independent — they are both manifestations of the same random thermal motion of carriers. A material with higher mobility also has a higher diffusion coefficient.

12.2 Hall Effect

The **Hall Effect**, discovered by E.H. Hall in 1879, is the phenomenon of development of a transverse voltage (Hall voltage, V_H) across a current-carrying conductor or semiconductor when a magnetic field is applied perpendicular to the direction of current flow.

12.2.1 Theory and Derivation

Setup: Consider an n-type semiconductor bar. Current I flows in $+x$ direction. Magnetic field B_z is applied in $+z$ direction.

Step 1: Electrons (moving in $-x$ direction) experience a Lorentz force:

$$F_L = q \cdot v_d \times B = (-q)(-v_d \hat{x}) \times (B_z \hat{z}) = qv_d B_z (\hat{x} \times \hat{z}) = -qv_d B_z \hat{y}$$

So electrons accumulate on the $-y$ face $\rightarrow$ Hall voltage develops.

Step 2: Equilibrium condition — electric force balances Lorentz force:

$$q \cdot E_H = q \cdot v_d \cdot B_z$$

$$E_H = v_d \cdot B_z$$

Step 3: Since $J_x = n \cdot q \cdot v_d$, so $v_d = J_x / (nq)$

Step 4: Hall voltage across width d :

$$V_H = E_H \cdot d = (J_x \cdot B_z \cdot d) / (nq)$$

Step 5: Hall coefficient R_H is defined as the ratio of Hall field to the product of current density and magnetic field:

Hall Coefficient and Related Formulas:

$$R_H = E_H / (J_x \cdot B_z) = 1 / (n \cdot q) \quad [\text{for n-type, } R_H < 0]$$

$$R_H = 1 / (p \cdot q) \quad [\text{for p-type, } R_H > 0]$$

$$V_H = R_H \cdot (I \cdot B) / t \quad (t = \text{thickness of sample})$$

$$\text{Hall mobility: } \mu_H = |R_H| \cdot \sigma = |R_H| / \rho$$

$$\text{Carrier concentration: } n = 1 / (R_H \cdot q)$$

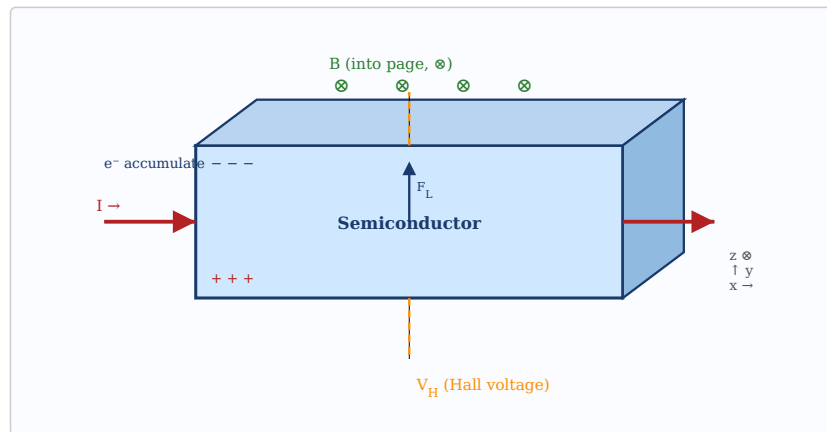


Fig. 12.1 – Hall Effect Setup: Current I along x , B along $z \rightarrow$ Hall voltage V_H along y

12.3 Applications of Hall Effect

- **Determination of carrier type:** Sign of V_H reveals whether the material is n-type (negative) or p-type (positive).
- **Measurement of carrier concentration:** n or p can be calculated from R_H .
- **Measurement of magnetic field:** Hall sensors are used in digital compasses, current sensors, and position sensors.
- **Hall effect multiplier:** Used in analog multiplier circuits ($V_H \propto I \times B$).
- **Speed sensors:** Used in brushless DC motors and automotive wheel speed sensors.
- **Gaussmeters:** Instruments to measure magnetic flux density.

Table 12.1 – Hall Effect Parameters Summary

Parameter	n-type	p-type
Majority carrier	Electrons	Holes
Hall coefficient R_H	$-1/(nq)$ (negative)	$+1/(pq)$ (positive)
Hall voltage V_H polarity	$-y$ face is negative	$-y$ face is positive
Hall mobility μ_H	$ R_H \cdot \sigma$	$ R_H \cdot \sigma$
Use	Identify n-type material	Identify p-type material

13. Questions & Answers

2-Mark Questions

Q1. Define Fermi level and state its position in an intrinsic semiconductor.

2 Marks

Fermi Level: The Fermi level (E_F) is defined as the energy level at which the probability of finding an electron is exactly 0.5 (50%) at any finite temperature. It is a reference energy that characterizes the statistical distribution of electrons.

Position in Intrinsic Semiconductor: In an intrinsic semiconductor at absolute zero, the Fermi level lies exactly at the middle of the forbidden energy gap, i.e., $E_F = (E_C + E_V)/2$. This is because the number of electrons in the conduction band equals the number of holes in the valence band ($n = p = n_i$), ensuring symmetric occupation of states.

Q2. What is Hall Effect? Write the formula for Hall coefficient.

2 Marks

Hall Effect: When a current-carrying semiconductor (or conductor) is placed in a magnetic field applied perpendicular to the direction of current flow, a transverse electric field (Hall field) and a corresponding voltage (Hall voltage) are developed across the sample in a direction perpendicular to both current and magnetic field.

This phenomenon was discovered by Edwin H. Hall in 1879.

Hall Coefficient: $R_H = E_H / (J \cdot B) = 1/(nq)$ for n-type; $R_H = 1/(pq)$ for p-type.

For n-type: R_H is negative. For p-type: R_H is positive.

Q3. Define mobility and write its unit. How does it vary with temperature?

2 Marks

Mobility (μ): The mobility of a charge carrier is defined as the magnitude of the drift velocity acquired per unit electric field. It measures how quickly a carrier moves through the semiconductor in response to an applied field.

$$\mu = v_d / E \quad (\text{Unit: m}^2/\text{V}\cdot\text{s} \text{ or cm}^2/\text{V}\cdot\text{s})$$

Temperature dependence: At low temperatures, mobility is limited by ionized impurity scattering and increases with temperature as $\mu \propto T^{(3/2)}$. At higher temperatures, lattice (phonon) scattering dominates and mobility decreases as $\mu \propto T^{(-3/2)}$. Overall, in most semiconductors at room temperature, mobility decreases with increasing temperature.

Q4. State Einstein's relation and its physical significance.

2 Marks

Einstein's Relation: The diffusion coefficient D and mobility μ of charge carriers are related by:

$$D/\mu = kT/q = V_T \quad (\text{Thermal voltage})$$

where k = Boltzmann's constant, T = absolute temperature, q = charge of electron.

Physical Significance: It shows that drift (mobility) and diffusion arise from the same physical process — random thermal motion of carriers. A semiconductor with high carrier mobility also has a high diffusion coefficient. At 300 K, $kT/q \approx 26$ mV. This relation is essential for designing p-n junctions and bipolar transistors.

5-Mark Questions

Q5. Explain the formation of energy bands in crystalline solids. Describe the classification of solids on the basis of energy bands.

5 Marks

Formation of Energy Bands: In a single isolated atom, electrons occupy well-defined, discrete energy levels governed by quantum mechanics. When N atoms are brought together to form a crystal, the electrons of neighbouring atoms interact with each other. According to the Pauli exclusion principle, no two electrons in a system can have the same set of quantum numbers. Therefore, the discrete energy levels of isolated atoms must split into N closely spaced sub-levels to accommodate all electrons.

Since $N \approx 10^{23}$ atoms/cm³ in a solid, the N sub-levels are so closely spaced that they effectively form a continuous **energy band**. The range of energies within a band is called the **allowed energy band**. Between adjacent allowed bands, there are ranges of energy for which no quantum states exist — these are called **forbidden energy gaps** or **band gaps**.

The two most important bands are: (a) **Valence Band (VB)** — the highest energy band that is completely or partially filled at 0 K, containing the outermost (valence) electrons; and (b) **Conduction Band (CB)** — the next higher energy band, which is empty or partially filled.

Classification Based on Energy Bands:

1. Conductors (Metals): The conduction band and valence band overlap, meaning a large number of electrons are always free to conduct. $E_g = 0$. Resistivity: $10^{-8} - 10^{-6} \Omega\cdot\text{m}$. Example: Cu, Ag.

2. Insulators: The valence band is completely filled and the conduction band is completely empty, separated by a very large band gap ($E_g > 3$ eV). Electrons cannot be thermally excited to the CB. Resistivity: $10^8 - 10^{16} \Omega\cdot\text{m}$. Example: Diamond ($E_g = 5.4$ eV).

3. Semiconductors: The valence band is full at 0 K and the CB is empty, but the band gap is small ($E_g = 0.1 - 3$ eV). At room temperature, thermal energy excites some electrons to the CB, creating electron-hole pairs. Resistivity: $10^{-4} - 0.5 \Omega\cdot\text{m}$. Example: Si ($E_g = 1.12$ eV), Ge ($E_g = 0.66$ eV).

Q6. Differentiate between intrinsic and extrinsic semiconductors. Explain n-type and p-type semiconductors with energy band diagrams.

5 Marks

Intrinsic Semiconductor: A pure semiconductor without any added impurities. Its conductivity depends only on temperature. Equal numbers of electrons and holes are generated thermally: $n = p = n_i$. Fermi level lies at the midgap. Example: Pure Si, Pure Ge.

Extrinsic Semiconductor: A semiconductor doped with controlled amounts of impurity atoms to enhance conductivity. The carrier concentration is primarily determined by doping. $n \neq p$ in general. Fermi level shifts from midgap depending on dopant type.

n-type Semiconductor: Formed by adding pentavalent impurities (P, As, Sb) to Si or Ge. Each dopant atom contributes one free electron to the crystal. These donor atoms create energy levels (donor levels E_D) just below the conduction band edge. The donated electrons are easily excited into the CB. Majority carriers = electrons; Minority carriers = holes. Fermi level moves closer to E_C . Carrier concentration: $n \approx N_D$, $p = n_i^2/N_D$.

p-type Semiconductor: Formed by adding trivalent impurities (B, Al, Ga) to Si or Ge. Each dopant atom has only 3 valence electrons and creates a vacancy (hole) in the covalent network. Acceptor levels (E_A) form just above the valence band. Electrons from the VB fill these levels, creating holes in the VB. Majority carriers = holes; Minority carriers = electrons. Fermi level moves closer to E_V . Carrier concentration: $p \approx N_A$, $n = n_i^2/N_A$.

Law of Mass Action (both types): $n \cdot p = n_i^2 = \text{constant}$ at a given temperature.

Q7. Distinguish between drift current and diffusion current. Derive the expressions for both in a semiconductor.

5 Marks

Drift Current: This is the current produced by the motion of charge carriers under the influence of an applied electric field. When a field E is applied, electrons drift in the $-E$ direction and holes drift in the $+E$ direction. The drift velocity is $v_d = \mu E$. The current densities are: $J_{n,drift} = nq\mu_n E$ (electrons) and $J_{p,drift} = pq\mu_p E$ (holes). Total drift current density: $J_{drift} = q(n\mu_n + p\mu_p)E = \sigma E$.

Diffusion Current: This arises due to the non-uniform spatial distribution of charge carriers. Carriers diffuse from regions of high concentration to regions of low concentration. The driving force is the concentration gradient dn/dx or dp/dx . Using Fick's law: $J_{n,diff} = qD_n(dn/dx)$ and $J_{p,diff} = -qD_p(dp/dx)$, where D_n and D_p are the electron and hole diffusion coefficients respectively.

Total Current: In a real semiconductor, both currents coexist. $J_n = qn\mu_n E + qD_n(dn/dx)$ and $J_p = qp\mu_p E - qD_p(dp/dx)$. These are the fundamental transport equations for semiconductor devices.

Key Difference: Drift needs an external electric field; diffusion needs only a concentration gradient. Both contribute to the total current in devices like p-n junctions and solar cells.

Q8. Derive Einstein's relation connecting diffusion coefficient and mobility.

5 Marks

Derivation of Einstein's Relation:

Consider a non-uniformly doped n-type semiconductor in thermal equilibrium. Even though there is no net current flow, a built-in electric field E exists due to the doping gradient. In equilibrium, the drift and diffusion currents exactly balance:

$$J_{n,total} = qn\mu_n E + qD_n(dn/dx) = 0 \quad \dots(i)$$

In thermal equilibrium, the electron concentration is given by Boltzmann statistics:

$$n(x) = n_0 \cdot \exp(q\phi(x)/kT) \quad \text{where } \phi \text{ is the local electrostatic potential}$$

$$\text{Differentiating with respect to } x: dn/dx = n \cdot (q/kT) \cdot (d\phi/dx)$$

$$\text{Since the electric field } E = -d\phi/dx, \text{ we get: } dn/dx = -nqE/kT \quad \dots(ii)$$

$$\text{Substituting (ii) into (i): } qn\mu_n E + qD_n(-nqE/kT) = 0$$

$$\text{Dividing throughout by } qnE: \mu_n - D_n q/kT = 0$$

$$\text{Therefore: } D_n/\mu_n = kT/q = V_T$$

$$\text{Similarly for holes: } D_p/\mu_p = kT/q = V_T$$

$$\text{At } T = 300 \text{ K: } V_T = kT/q = (1.38 \times 10^{-23} \times 300)/(1.6 \times 10^{-19}) \approx 0.026 \text{ V} = 26 \text{ mV}$$

Significance: The relation shows that mobility and diffusion are intimately connected through thermal energy kT . It is used extensively in the analysis of p-n junctions, minority carrier transport, and bipolar transistors.

Q9. Explain the Hall Effect in detail. Derive an expression for Hall coefficient and mention its applications.

5 Marks

Hall Effect: When a semiconductor (or conductor) carrying a current I in the x -direction is placed in a magnetic field B applied in the z -direction, a transverse electric field (Hall field E_H) is developed in the y -direction. This is called the Hall Effect.

Derivation of Hall Coefficient:

For an n-type semiconductor: electrons move in $-x$ direction (conventional current in $+x$). When B is in $+z$ direction, the Lorentz force on electrons is: $F = q(v \times B)$. For electrons: $F_L = (-q)(-v_d \hat{x}) \times (B \hat{z}) = -qv_d B \hat{y}$. Electrons accumulate on the $-y$ face.

This charge buildup creates an electric field E_H in the $-y$ direction (for n-type, $-y$ face becomes negative). At equilibrium: $qE_H = qv_d B$, so $E_H = v_d B$.

Since $J_x = nqv_d$, we have $v_d = J_x/(nq)$.

Therefore: $E_H = J_x B/(nq)$.

The Hall coefficient: $R_H = E_H/(J_x B) = 1/(nq)$ for n-type (negative for electrons) and $R_H = 1/(pq)$ for p-type (positive for holes).

Hall voltage: $V_H = E_H \cdot d = (I \cdot B)/(nqt)$ where d = width, t = thickness.

Applications: (1) Determine whether semiconductor is n-type or p-type by sign of V_H . (2) Measure carrier concentration n or p . (3) Measure magnetic field strength (gaussmeters, magnetic sensors). (4) Hall-effect current sensors (contactless current measurement). (5) Position sensing in brushless motors and automotive systems. (6) Hall-effect multiplier circuits.

14. Quick Revision Sheet

⚡ Key Formulas

- $n_i = \sqrt{N_c N_v} \cdot \exp(-E_g/2kT)$
- $\sigma = q(n\mu_n + p\mu_p)$
- $J_{\text{drift}} = \sigma E = q(n\mu_n + p\mu_p)E$
- $J_{n,\text{diff}} = qD_n(dn/dx)$
- $J_{p,\text{diff}} = -qD_p(dp/dx)$
- $D/\mu = kT/q = V_T \approx 26 \text{ mV}$ (Einstein)
- $R_H = 1/(nq)$ [n-type] = $1/(pq)$ [p-type]
- $n \cdot p = n_i^2$ (Law of Mass Action)
- $V_H = R_H \cdot IB/t$

◆ Band Theory Summary

- N atoms $\rightarrow$ N sub-levels $\rightarrow$ energy band
- Conductor: $E_g = 0$ (overlap)
- Semiconductor: $E_g = 0.1\text{-}3 \text{ eV}$
- Insulator: $E_g > 3 \text{ eV}$
- Si: $E_g = 1.12 \text{ eV}$; Ge: $E_g = 0.66 \text{ eV}$
- Fermi-Dirac: $f(E) = 1/[1 + \exp((E - E_F)/kT)]$
- $n \cdot p = n_i^2$

⚡ Carrier Transport

- Drift: due to electric field E
- $v_d = \mu \cdot E$
- Diffusion: due to concentration gradient
- Both occur simultaneously in devices
- Total $J_n = qn\mu_n E + qD_n dn/dx$
- Total $J_p = qp\mu_p E - qD_p dp/dx$

● Fermi Level Concepts

- Intrinsic: $E_F \approx \text{midgap}$
- n-type: E_F closer to E_C
- p-type: E_F closer to E_V
- High T: $E_F \rightarrow E_i$ for all types
- High doping $\rightarrow E_F$ enters band (degenerate)

● Hall Effect Key Points

- Current I in x; B in z $\rightarrow V_H$ in y
- n-type: R_H negative ($-1/nq$)
- p-type: R_H positive ($+1/pq$)
- Gives: carrier type, n or p, μ_H
- Uses: sensors, gaussmeters, ICs

📄 Semiconductor Values (Si at 300 K)

- $n_i \approx 1.5 \times 10^{10} \text{ cm}^{-3}$
- $\mu_n = 1350, \mu_p = 480 \text{ cm}^2/\text{V}\cdot\text{s}$
- $E_g = 1.12 \text{ eV}$
- $kT/q = 26 \text{ mV}$ at 300 K
- $D_n = 35 \text{ cm}^2/\text{s}, D_p = 12.5 \text{ cm}^2/\text{s}$

Master Formula Sheet - Unit V Semiconductors

$$n_i = \sqrt{N_C N_V} \cdot \exp(-E_g/2kT) \quad [\text{Intrinsic concentration}]$$

$$\sigma = q \cdot (n \cdot \mu_n + p \cdot \mu_p) \quad [\text{Conductivity}]$$

$$J = nq\mu E \quad [\text{Drift current density}]$$

$$J_{\text{diff}} = qD \cdot (dn/dx) \quad [\text{Diffusion current density}]$$

$$D/\mu = kT/q \quad [\text{Einstein's relation}]$$

$$R_H = 1/(nq) \quad [\text{Hall coefficient, n-type}]$$

$$n \cdot p = n_i^2 \quad [\text{Law of Mass Action}]$$

$$E_F(\text{intrinsic}) = (E_C + E_V)/2 \quad [\text{Fermi level}]$$

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End of Unit V - Semiconductors

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